

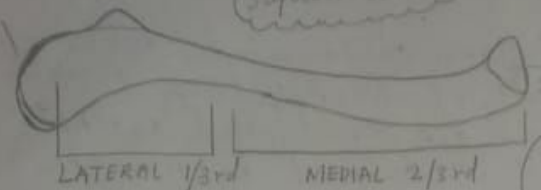
## Upper Limb

## Chp 2 - Bones

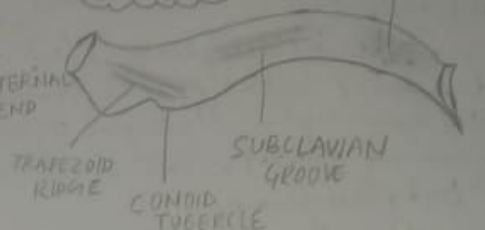
### Clavicle

ACROMIAL END

Superior View



Inferior View



### Disruption

- △ clavicle is a long bone
- △ also called as collar bone/beauty bone
- △ supports the shoulder so that arm can swing clearly away from trunk.
- △ parts → shaft (curved)
  - lateral end or acromial
  - medial end or sternal

### Side Determination

- △ shaft is slightly curved
- △ lateral end is flattened
- △ medial end is quadrilateral
- △ medial side is convex forward (2/3rd)
- △ lateral end is concave forward (1/3rd)
- △ inferior surface is grooved longitudinally (subclavian groove)
  - △ has trapezoid ridge on lateral side.

### Peculiarities

- △ only long bone that lies horizontally.
- △ Subcutaneous throughout
- △ it has no medullary cavity.
- △ it is 1st bone to start ossify
- △ It is the only long bone which has two primary centres of ossification.

### Features

- △ shaft: is divisible into medial 2/3rd & lateral 1/3rd.
- △ Lateral 1/3rd: △ flattened
  - △ it has 2 borders Anterior & posterior concave & convex forward.
- △ it has 2 surfaces Superior & inferior
  - △ Superior → Subcutaneous
  - △ Inferior → Conoid tubercle & Trapezoid Ridge

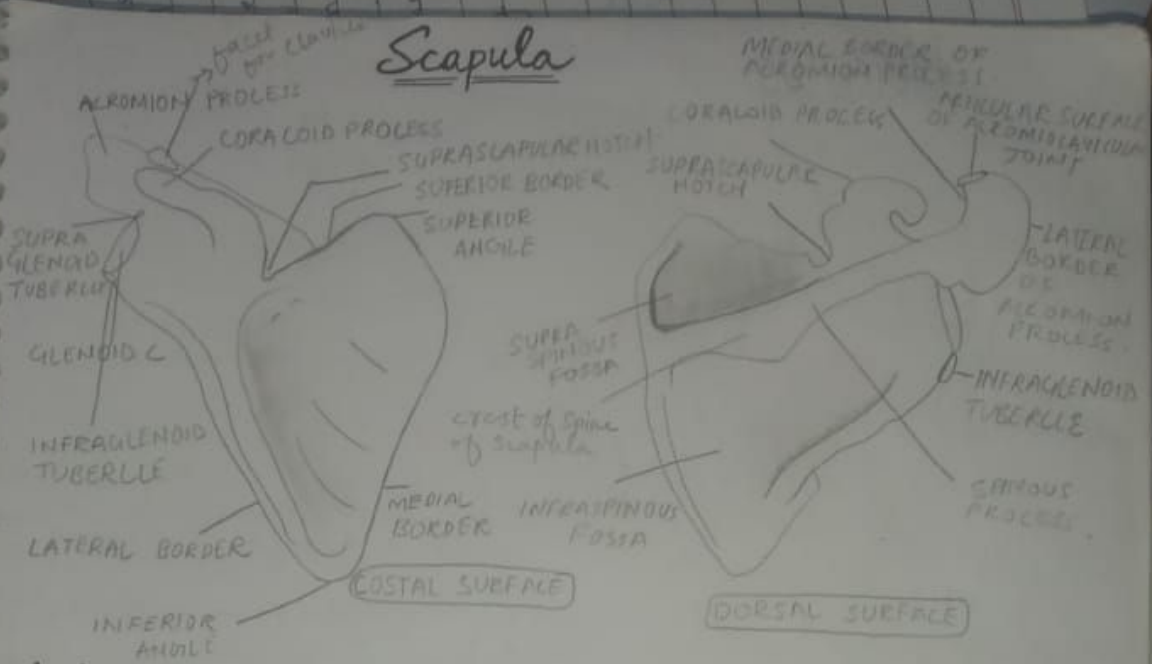
(Elevations are for ligament Attachment)

- △ Articulation with Acromion of Scapula [Acromioclavicular joint]
  - medial 2/3rd.
- △ Quadrangular in shape
- △ 4 Surfaces →
  - Anterior: convex, posterior: concave
  - Superior: Rough for muscle Attachment (Subcutaneous)
  - Inferior: subclavian groove.

### Clinical Anatomy

- △ Congenitally Absent of clavicle called cleidocranial dysostosis.
- △ clavicle is commonly fractured bone

begin all deficit surgery is over leg one is sent in apex



### Disuption

- called as **shoulder blade**
- It is thin bone
- posterolateral aspect of Thoracic cage
- Scapula has → 2 Surfaces  
→ 3 borders  
→ 3 angles  
→ 3 processes

### Side Determination

1. **Lateral (Glenoid) Angle** is large angle and bears glenoid cavity.
2. **Thickest lateral border** runs from the Glenoid cavity (above) to the inferior angle (below).
3. **Dorsal Surface**
  - △ Convex in shape
  - △ Divided by triangular spine into supraspinous fossa & infraspinous fossa.
4. **Costal Surface**
  - △ Concave in shape
  - △ Occupied by Subscapular fossa fixed into concave chest wall.

### Features

#### Surfaces:-

1. Dorsal Surface
2. Costal Surface. [Thick ridge at lateral border, this part of bone is almost rod-like. It acts as lever for action of serratus anterior in overhead Abduction of Arm.]

#### Borders:-

1. Superior Border is Shortest, Near the root of coracoid process, suprascapular notch is also present.
2. Lateral Border is thick and starts from infraglenoid tubercle to inferior angle.
3. Medial Border is thin. It extends from Superior Angle to Inferior Angle.

#### Angles:-

1. Superior Angle
2. Inferior Angle
3. Lateral Angle Bears Glenoid cavity or fossa.

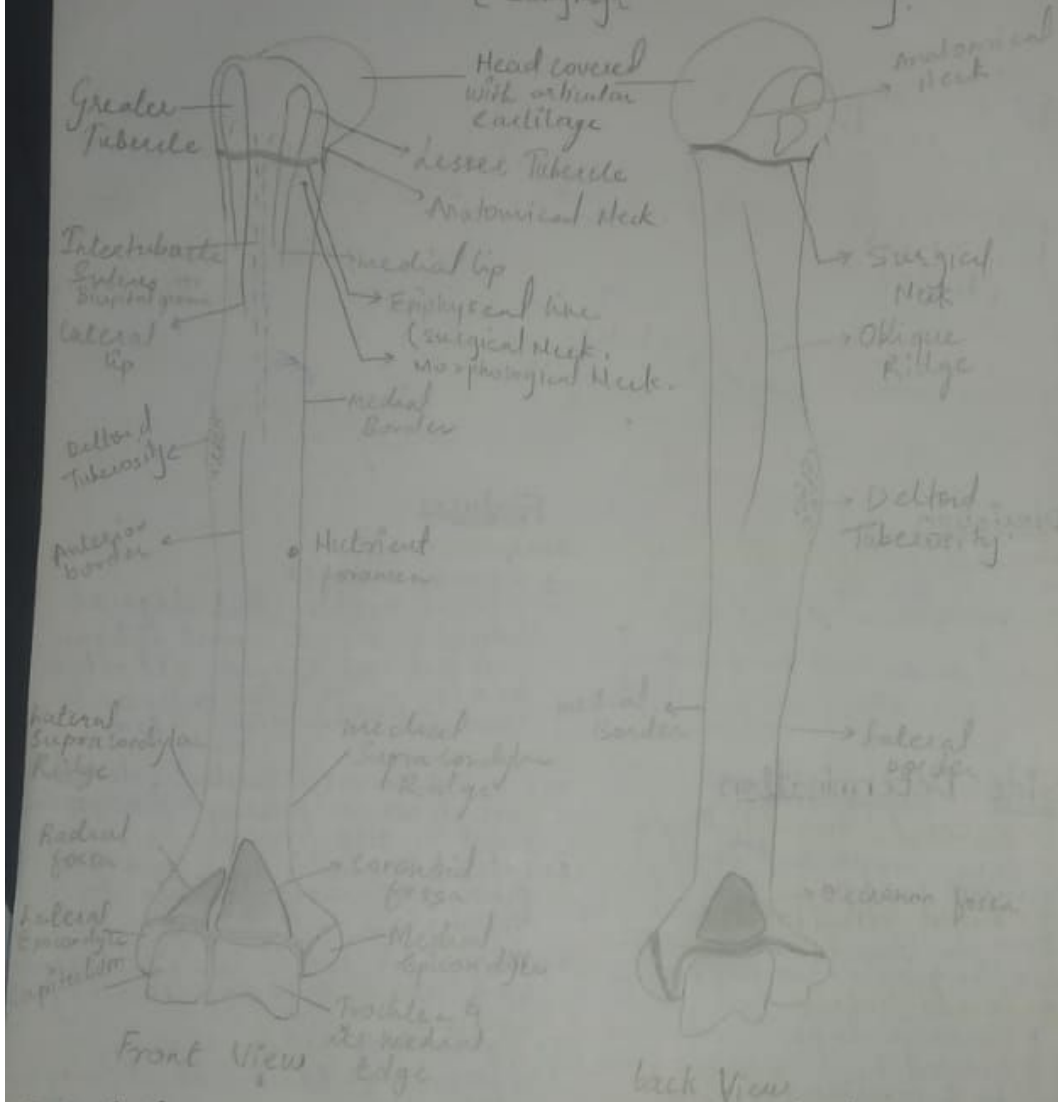
#### processes:-

1. Spine or Spinous processes
2. Acromion processes has 2 borders, medial & lateral, 2 Surfaces sup & inf and facet for clavicle.
3. Coracoid process → finger like

Clinical Anatomy paralysis of Serratus Anterior Causes Winging of Scapula.

# HUMERUS

[3 Hark ke baare mein  
Features mein acche se  
Samjhega]



## Description

- It is the bone of Arm
- longest bone of Upper limb
- Divisible into Upper end & Shaft.

## Side Determinations

- Upper End is directed medially, upward & backward
- and rounded to form head
- Lesser tubercle is located in front of Upper end and
- greater tubercle is located in lateral side
- lower end is directed from side



## PARTS

### UPPER END

- 1: Head: Smooth Rounded
- 2: Neck
- 3: Greater Tubercle
- 4: Lesser Tubercle
- 5: Intertubercle Sulcus

### SHAFT

Upper half: Rounded

Lower half: Triangular

### LOWER END

- 1: Capitulum (lateral small process projection)
- 2: medial & pulley shape Trochlea
- 3: Coronoid fossa Above Trochlea
- 4: Radial fossa Above Capitulum
- 5: medial & lateral Epicondyle
- 6: Olecranon fossa (Deep hollow on posterior aspect)

## FEATURES

### \* UPPER END

→ Head: Smooth, round,  $\frac{1}{3}$ rd of sphere. Covered by hyaline cartilage

→ Neck

→ **ANATOMICAL**: Line separates the head from the rest of upper end.

→ **SURGICAL**: • Narrow line / Epiphyseal line separating upper end from the shaft.  
• most common site of humerus fracture.

→ **MORPHOLOGICAL**: lies just below the surgical neck (0.5 cm).

→ **GREATER TUBERCLE**: most lateral part of proximal end, upper end.

→ **LESSER TUBERCLE**: Small Elevation on front upper end.

→ **Intertubercle Sulcus**: Bicipital Groove

### \* LOWER END

- △ Capitulum articulate with head of Radius to form elbow joint.
- △ Trochlea articulate with trochlear notch of Ulna
- △ Coronoid process of Ulna meets coronoid fossa of humerus during full Elbow flexion
- △ Head of Radius meets Radial fossa of humerus during full Elbow flexion
- △ Olecranon process of Ulna meets Olecranon fossa of Humerus during full Elbow Extension.

### \* SHAFT

→ Upper part is cylindrical  
→ lower part is triangular

**3 Surfaces & 3 Borders**

→ Anterolateral: V-shaped Deltoid Tuberosity

→ Anteromedial: Nutrient Foramen

→ posterior: Upper  $\frac{1}{3}$ rd Oblique Ridge directing downward and laterally

→ Anterior: From lateral tip of intertubercular Sulcus to Ant. margin of Deltoid Tuberosity and ends in Radial Fossa.

→ medial: medial supracondylar Ridge

→ lateral: lateral supracondylar Ridge

## Clinical Anatomy

### Common Fracture Sites of Humerus.

- Surgical Neck
- Shaft
- Supra Condylar Region

### SUPRALONDYLAR FRACTURE

- ▷ Common in children & youngsters
- ▷ Usually happens when you fall on an outstretched hand.
- ▷ Lower fragment of broken bone usually shifts backward & making the Elbow look more prominent (Sign of fracture).
- ▷ Median nerve can be damaged / injured
- ▷ can also block brachial Artery.

### How to Detect Elbow Fracture or Dislocation

- ▷ When the Elbow is bent (flexed), The Olecranon (Elbow tip) and the two Epicondyle form an Equilateral triangle.
- ▷ When the Elbow is straight (extended) These points lie on a straight line.
- ▷ These Relationships are important for detecting Elbow Dislocations & Elbow fractures.

### POOR BLOOD SUPPLY AREA

- ▷ The junction of upper and middle Thirds of Humerus
- ▷ Fractures fail to unite or heal slowly.

Anatomical Neck.

### HEAD DISLOCATION

- ▷ Head of Humerus usually dislocate Downwards (Inferiorly)
- ▷ This is because of lower part of Shoulder joint Capsule is weak.